

Coupling Services

Enabling Technologies to Cross the Venture Divide

Business Proposal

Executive Summary
Condensed Business Plan
Appendices

by

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Board of Advisors

We would like to begin by thanking our distinguished Board of Advisors for their invaluable questions and feedback. Without their advice, the quality of our plan would not be what it is today. Others are currently considering joining our Board of Advisors.

Mr. Jamey Bennett – Mr. Bennett is a Bucknell University alum and currently an Entrepreneur in Residence at FrontierCapital LLC, a venture capital firm. He is a successful entrepreneur who has founded two groundbreaking Internet-related businesses during the past six years. His most recent endeavor was co-founding LendingTree (Nasdaq: TREE), a successful consumer loan marketplace, which completed its initial public offering in February 2000. Prior to LendingTree, he founded BookWire, one of the first “vertical B2B” web sites. BookWire was sold to Cahners Publishing in November 1996. Mr. Bennett currently serves on the boards of CommissionsToday.com and eMarketCapital.com

Mr. Thomas Rippon, Esq. – Mr. Rippon is a successful central Pennsylvanian entrepreneur and president of T. R. Rippon & Associates. He has experience as an angel investor and as a member of a local venture capital group. Although Mr. Rippon is a graduate of Colby, he is a supporter of the arts at Bucknell University.

Prof. Charles Moran, Esq. – Prof. Moran is a faculty member of the Management Department at Bucknell University. He teaches a Senior, capstone course entitled “Entrepreneurial Venturing”. His teaching does have a practical root though. Prior to fulfilling his life’s dream of teaching, Prof. Moran raised ten million dollars in funding to start his own closing corporation on Wall St. Prof. Moran was also a prominent member of several non-profit organizations. He currently teaches at Harvard’s Advanced Management Program during the summers.

Executive Summary

Coupling Services (CS) creates businesses by fusing high potential technologies with management expertise, product development, and capital that technology developers lack. Typically, developers are forced to enter licensing agreements with current corporations at a significant loss compared to research expenditures. Instead, CS will mold the support necessary for technology commercialization through a startup business. Institutions and universities whose technologies are incorporated in this manner will potentially recover a much larger percentage of money spent on research to fund future endeavors.

CS will perform a wide range of business development services, including producing written business plans for submission to Angel investors and Venture capitalists. Our services reduce investor risk levels by providing more feasible and profitable business models, thereby greatly improving the probability of a return on investments. We supply investors with experienced entrepreneurs and management, deep industry analysis, intense market research, and thorough technology assessment. Additionally, we enable universities to comply with the Bayh-Dole Act of 1980. This Act requires universities to favor small companies and startups over large corporations when commercializing or licensing technologies.

Initially, technologies are screened through a panel of internal and external experts to determine those technologies with strongest market potential and feasibility. Then an interdisciplinary team of experienced professionals conduct initial research of marketability, estimation of costs to market, profit potential, analysis of prospective risks, acquisition of needed management, as well as gathering other crucial information to create a business with a solid foundation. CS will utilize external consultants and industry experts as necessary to assess a technology's potential and the validity of its claimed usefulness. Having established the initial

framework for a successful company, CS will solicit angel investment in the future portfolio company. Our business model is not real estate based like most incubators.

CS fills a niche in the market that is currently not being filled. The current licensing environment favors established corporations at the expense of academic institutions. Many universities regularly accept licensing agreements, because they lack the funds and expertise to start a business. Please see the current university technology transfer model on page 31. Increasingly institutions prefer an equity stake in startups. CS joins our clients' interests with investors. In the long run, creating a new business with our services will increase both universities' and our own wealth, while increasing investors success rate. It's a win-win situation for technology developers, investors, and CS. Please see Coupling Services' technology transfer model on page 31.

CS will foster a diversified portfolio of technologies and a team-based structure that allows for collaboration, cross-fertilization and scalable growth. As more institutions desire our services, additional interdisciplinary teams will be added to the infrastructure. Currently, many institutionally developed technologies cannot be commercialized into a viable business individually and are "shelved" and forgotten. However, by synergistically combining several or many similar technologies, profitable businesses may be formed. With knowledge of all available technologies, CS will have the unique ability to use these types of technologies for commercialization. Our business model allows all costs to be passed along to investors and not universities. We make money by charging investors a fee for the technology plans they choose.

Profitability measures include a 46.9% IRR and a net present value of \$9,400,000 (7.5% discount rate) after five years in operation. This IRR does not include the potential sale of equity in portfolio companies. Currently, CS is seeking a \$200K initial angel investment for business model validation. Initial estimates project a \$3 million requirement in venture capital funding.

Business Concept

Opportunity

Hundreds of discoveries are made and technologies developed each year by individuals at research institutions. Most of these technologies are commercialized through the use of licensing agreements. According to the latest report from the Association of University Technology Managers (AUTM), 190 academic institutions contributed over seven billion dollars on research expenditures in 1999, yet received less than a billion dollars in licensing income.¹

Because technology developers lack the time, expertise, and capital to start their own businesses, they enter licensing agreements with large corporations at a significant loss. If technology developers were able to start their own businesses, they could potentially recoup a larger percentage of the money spent on research to fund future endeavors. Currently, established corporations increase their own revenues by limiting payments to patent owners and technology developers. For example, “Rob Johnson, a commercial analyst for Onyvax Ltd., says his company stands to save millions of dollars in research and development by buying technology already in existence. ‘We can’t justify spending large amounts on basic research. (It’s) easier, cheaper and less risky to license it in after someone else has done it.’”¹⁵ Please see our depicted business concept graphic on page 30.

The Bayh-Dole Act (Public Law 96-517) of 1980 requires universities and other nonprofit recipients of federal funding to favor small businesses and start-ups over large corporations, when determining who is allowed to own patents arising out of university

research.⁷ Unlike most entrepreneurs, universities prefer a non-controlling stake in start-up companies built upon their research. This eases negotiations and allows investors to share a larger equity stake, while providing the opportunity to raise additional revenues through supplementary equity offerings. Such offerings can be presented to institutions as a payment comparable to average licensing fees. Currently, we have only found Columbia University to be in full compliance with the act. This is because their technology licensing office actively attempts to create new businesses around university developed technology. As a case in point, Columbia receives the most revenue from their technologies when compared to all private universities in the U.S.¹⁸ CS will attempt to bring the type of success Columbia has seen to every research institution.

Forming new ventures often requires significant seed and/or venture capital. However, as a recent article in Business Week stated, "Many VCs took shortcuts during the Net frenzy, (losing millions on unsound Internet business models). They didn't check out entrepreneurs thoroughly. They didn't conduct deep industry research. And they didn't verify technologies." With the slowing economy, there is now, "likely to be a flight to quality."² CS, through our due diligence services, increases the quality of business models.

Company Description

CS' focus is to join university developers' high potential technologies with the funding and support needed to create their own businesses. The company will be based in a region supporting cutting-edge academic research. Coupling Services Inc. will provide due diligence and business development services including writing business plans for submission to angel investors and venture capitalists. By creating new ventures, more research dollars are potentially generated for universities and researchers when compared to current licensing revenue.

CS consists of a flat structure with interdisciplinary teams, including a support staff. Each team has engineering, business plan writing, marketing, finance, entrepreneurial, and business development skills. Additionally, teams have access to business and patent lawyers, industry and technology expertise, and executive recruitment resources. We provide investors with experienced entrepreneurs, intense market research, and thorough technology assessment. Our services reduce investor risk levels by improving the probability of a return on their investments and provide an incentive for universities to comply with the Bayh-Dole Act.

Services Provided

CS does much more than write business plans or act as a broker. We integrate the services needed to successfully start businesses. CS provides a variety of services enabling technology developers to start new businesses. CS screens technologies through a panel of internal and external experts. Based upon their evaluations, we determine candidates with strong potential that warrant further efforts in formulating a complete business plan. At this point, we request the research institutions postpone their efforts in pursuing a licensing agreement with outside corporations while we seek funding for a new business.

CS' team of experts handles initial researching of marketability, estimation of costs, analyzing potential risks, obtaining needed management, as well as gathering other crucial components to formulate a quality business plan and build a successful business. If angel/venture capital or other investments cannot be obtained, we will then seek licensing agreements. Our internal and external screening will provide value to the licensing process. For example, companies identified as potential competitors for the proposed venture would now become licensing targets if a new company could not be funded.

Our intent is not to turn professors into executive management. Rather, we carefully select an experienced management team and empower them to operate the business from the

beginning. To be clear, the recruited management team will be responsible for operating the companies we help to form, not us. Our services benefit researchers by allowing them to maintain their current obligations, do what they do best, have a voice in how their life's work will benefit society, and find support for their current academic efforts. Likewise, academic institutions have the opportunity to recoup more expenditure by eliminating corporate exploitation of their research and development efforts. In a venture world of incubators and seed capital, we are the pre-incubator fertilizer that nourishes the start-up for explosive growth.

At the same time, CS approaches angel investors and venture capitalists. It is less expensive for investors to pay us to do the needed research and confirmation than for them to screen and research all the plans they receive. Also, they only pay for these services when they feel the business opportunities/plans are worth funding. This concept is much like the profitable head-hunter/recruiter business. In short, CS will provide highly customizable services necessary to create profitable companies, while limiting the risks early-stage investors currently bare.

How We Will Make Money

Our expense/revenue structure is designed to share most of our business development costs with Angel investors who are enthusiastic about new business opportunities. If we cannot convince Angels of our value proposition, we will not pursue it further or incur any related expenses of further developing the proposed businesses. In return for their risk exposure, Angel investors will receive a future percentage of any licensing deals if the new businesses were to fail. CS will take a minor stake ($\approx 5\%$) in the companies it helps to form. Having a vested interest in the companies will create an added incentive to ensure their future success and the quality of our services.

Additionally, if we cannot find funding for the businesses from Angel or Venture Capitalists, we will investigate licensing opportunities. Having done much of the footwork in

developing a technology's business plan (estimating potential market size, identifying competitors, determining cost to market) will give us a distinct advantage when approaching licensors.⁶ To aid in our initial operations, we will select technologies far along in the development process. By favoring these types of technologies in the beginning, we decrease the time to market while increasing the probability of finding angel and venture investment.

It is important to note that if a company or technology is not successful, we are not paid. In addition, we do not rent space like an incubator. Incubators are paid regardless of the success of a company and are largely a real-estate company. Like an employment recruiter, we make money with a successful match between investors and technologies.

Marketing

Industry Overview

CS' proposed business model straddles two industries: technology transfer and business development. Organizations including existing companies, Federal labs, defense contractors, and academic institutions have technologies they would like to commercialize, or bring to market, through selling the rights to other corporations. There has been a recent outbreak of on-line brokers who try to match those who have technologies with those who need technologies. Some of the most notable on-line brokers include (<http://www.uventures.com>) and (<http://www.yet2.com>). They typically charge a fee for access to their database of contact information. Business development organizations include incubators and service providers, such as business plan writing firms. There are an increasing number of incubators with specific ties to academic institutions, and several are actually located in university affiliated research parks.

Market Description

In 1993, 285 new businesses were created based on academic institutions' research and development. By 1999, this number had grown to 344. Assuming this type of continued growth, our potential market share will be roughly 46% by year five. We have not inflated our

projections to include potential growth of the market size resulting from our services. "Hundreds of universities have consistently written well over 3,000 new license agreements every year for the past three years, 3,914 in fiscal year 1999 alone, and over ten percent of these new agreements have consistently been with Start-ups."¹ Since 1980, at least 2,922 new companies have been formed based on a license from an academic institution, including the 344 reported in 1999. While many of these companies are younger than five years old, 70% of all companies started to-date remain operational today.¹

James Severson, past President of AUTM, said, "In many cases the only path for commercial development was to form these new companies specifically for the development of the technology. These results reflect that the academic research enterprise can provide the raw materials for development by small companies and that there is increased interest in academic institutions as a source of new ideas and technologies for development."¹ Corporate research and development tends to be more incremental in nature and risk adverse. Fiscal year 1999 results show that at least 417 new products were introduced from 98 institutions. On average, licensing royalties make up only 5% of total product sales.

Increasingly, institutions accept an equity stake in new businesses instead of licensing royalties. In fiscal year 1999, academic institutions received an equity interest in 243 transactions.¹ "Universities now routinely take equity in start-up companies, although such equity is normally a sweetener accounting for a small portion of the value associated with a technology license. Few universities 'start' companies and even fewer have the funds to do so on their own."³ Willingness on the part of academic institutions to foster creation of new companies and by faculty inventors to license their inventions to start-up companies is important for our business model. Frequently, this approach is the only way to develop very

forward-looking ideas. Existing companies are reluctant to invest heavily in ideas with uncertain outcomes.¹² According to the Council on Governmental Relations, “Many early stage technologies need a ‘champion’ for successful transfer and would languish but for the focused efforts of startup companies.”¹⁷

Questionnaire

We are currently awaiting responses to an e-mail survey mailed to 118 Technology Licensing Offices in North America. Ed Reilly really is writing a thesis about technology transfer, so we hoped to kill two birds with one stone with the survey. From personal interviews, we have learned that Licensing Offices are under funded and under staffed. To make our beliefs fact, we asked question one about their office resources. To see a copy of the questionnaire, please see page 32. Since the Bayh-Dole Act requires universities to favor small businesses in bidding, we asked question two to see how often they actually have to choose a small company. In a personal interview with Karen Thornton, Associate Director of the Hinman’s Entrepreneurship program at Univ. of Maryland, she said that Maryland’s licensing office had “cabinets full” of technologies waiting to be looked through. In question three, we wanted to see how many other schools had cabinets full of technologies awaiting licensing as well. In question five, we are specifically asking if there is a need for Coupling Services. Finally, in question six, we hoped to learn the most important factors schools use in choosing service providers.

Competitor Analysis

Angle Technology Limited (ATL) is a new and small player to the tech-transfer, business development field and our closest competitor. Started in April of 1994, they are based in the United Kingdom, have 19 employees, and received funding from both venture capital (GBP 3.2 million) and the British Government (GBP 400,000). Their goal is to license or create start-ups with corporate, government, and academic based technologies in the hopes of creating new jobs

within the United Kingdom. They also provide a variety of consulting services, including strategic planning for research parks.⁸ In October of 2000, ATL established a presence in the U.S. but is only working with the University of Virginia. ATL's business model for commercializing technology is not apparent. After eight years, they have only four portfolio companies to their credit, three biotech, and one surface chemistry technology.

Research Corporation Technologies (RCT) and Competitive Technologies, Inc. (CTT) are the only other U.S. competitors positioned for relatively easy entry to market. However, RCT and CTT's business models and fundamental strategies differ greatly from that of CS. Although RCT and CTT claim they assist universities start new companies, they did not help create any new companies in 1999.¹ Clearly, their strategic emphasis is not on building profitable businesses, but rather on licensing technologies to current corporations. Also, they do not foster a diversified portfolio of technologies in their licensing efforts.

RCT's business model centers on licensing developing biotechnology, while CTT licenses life and physical science technologies. By concentrating their efforts on this narrow market segment, RCT and CTT increase the level of risk to their investors and do not take advantage of a diversified portfolio of technologies methodology. Initially, CS will not limit its efforts to biotechnology. The development of biotechnology companies is highly capital intensive, while any return "commonly requires seven to twelve years" to materialize.¹⁰ These large and costly development constraints for biotechnology are typical for large, established corporations as well.

Other potential competitors include investment banks, incubators, and venture capital firms. For a large upfront fee, investment banks will write a business plan. However, their compensation is not contingent on technology developers successfully receiving funding, and universities rarely (if ever) choose this option. Although incubators and venture capital firms

have a low barrier to entry to this market, we hope to be the first to obtain capital for tech developers in starting their own businesses. Thus, we will be the first to have recognized success and the resulting credibility. Additionally, incubators and venture capital firms generally wait for opportunities to approach them. Most incubators are only aware of local or regional opportunities and have a profitability model based in real estate.

It is true that angel investors and venture capitalists could do much of what we propose on their own, but the examples we have found have been regionally confined in scope, such as, Redleaf Capital in Philadelphia and Pittsburgh. We propose to track developing technologies at over one hundred research institutions in North America. The real fifty thousand dollar question is why don't investors just outsource the various services we propose independently and avoid paying our fee? The answer is twofold. First, we integrate our services under one roof. We believe creating a new business is a collaborative affair best suited for a face-to-face exchange of ideas and information. Second, we screen numerous technologies to find the most attractive opportunities. If one chooses to outsource, they'll need to screen as many technologies as we do to find the "one" worth investment.

Lastly, a number of much smaller firms exist with extremely specific competencies. Their small size, typically 2-5 individuals, does not allow them to provide the number or breadth of services that Coupling Service will provide. These smaller firms will most likely never achieve the critical mass necessary for large market penetration. In many instances, CS may consider creating alliances with these firms, as they can provide specific expert evaluation. We realize at this time these small boutique firms are actually larger than we are right now, but we are still in the business development stage. What we propose to do is on a larger scale that won't be limited to one industry such as these firms typically are.

Competitive Advantages

The current licensing environment favors established corporations at the expense of academic institutions. As we have said, corporations profit by paying less to technology developers. The focal point of CS' efforts is our clients' interests. Unlike corporate middlemen, we are invested in academic institutions' success. We offer an attractive alternative to licensing by enabling technology developers to reap the benefits of their work through the creation of new businesses, when appropriate, and generate greater revenue to recover more of their costs.

- Focus on creating business ventures with university technology fills an underdeveloped niche market.
- Enable universities to comply fully with the Bayh-Dole Act.
- Provide investors with well-refined business opportunities that reduce their risk and save them time and money in performing their own extensive due diligence
- Team based approach allows easy scalability and adaptability for growth while providing greater, specific, expertise
- Ability for integration of proposed companies' design and goals.

Currently, many developers believe they have no option but to license their technologies to large corporations. Many times, these technologies are developed through corporate research agendas imposed on universities.⁵ CS will not impose research topics on academic institutions. We believe imposed research agendas hinder innovation. Also, there exists a wealth of promising technologies awaiting commercialization, without these constraints on academic freedom as evident by the number of available technology lists online.

- We present technologies to Angel investors they would not typically see; Academic institutions do not actively seek out private investors.

- Actively investigating and pursuing university technologies would be a dramatic change from the way incubators and venture capital firms currently conduct business.
- Emphasis on building relationships with many U.S. universities and investors will enable CS to quickly start-up many more companies and to fend off Angle Technology's entry.
- Since ninety percent of licenses and options to start-ups were exclusive in 1999,¹ there is a barrier to entry for their established corporate competitors.

Unlike our competitors, such as Angle Technologies and RCT, one of our goals is to have a portfolio of diverse technologies, an approach similar to the diversification efforts of many mutual and venture capital funds. By not limiting ourselves to one or two particular industries, we minimize potential losses from uncertain market conditions and make ourselves a more attractive investment compared to the competition. Other companies have unsuccessfully tried similar business models to what we propose, but there typically remains one big difference. We outsource the funding of new businesses to reduce our risk exposure. Those who fund new technology businesses must wait long periods to see any rewards. Our business model provides milestones for success by encouraging Angel investors to fund initial business development while receiving rights to some future licensing percentage if the new business fails. If some success is obtained, then we would encourage venture capital to step in and make a large investment.

Marketing Plans

There are three main market segments CS will need to consider: Academic Institutions, Faculty inventors, and investors. Below, we describe how we will target each segment. These plans follow logically from our competitive advantages.

Through discussions with academic technology licensing offices, we have found that these offices look for the easiest way to recover a portion of their research expenditures.

Inevitably, the method of choice has been through licensing agreements with large corporations.^{1,4} Universities and their technology licensing offices recover a tiny fraction of their research expenditures. Many believe it is beyond their reach to take part in forming new ventures. From the beginning, our priority will be to alter this perception through direct marketing and relationship building.

In effect, CS will change the status quo by making business formation the easiest solution for money recovery. Authorities with a vested interest in licensing have told universities for years why they cannot “cross the venture gap” and start up companies.¹³ We provide the solutions (the expertise, verification, research, capital, and experienced management) for these proposed problems. We will take on a symbiotic relationship while marketing to the technology developing market segment.

Even though licensing technologies to start-ups can generate more revenue in the long run to support research and encourage economic development in the region, our market research indicates that universities and faculty may not be excited about giving up the immediacy of licensing fees. Since universities typically do not want a controlling stake of new businesses founded on their research, CS will offer additional equity to investors in return for supplementary funds to be offered to institutions as an initial, lump-sum licensing fee, over-and-above an equity stake in the start-ups.

Our marketing strategy for the investor segment will be direct sales through networking and relationship development with Angels, Venture Capitalists, and their referring service providers. In our estimated expenses, we have provided extensive resources for travel to meet with technology developers, their institutions, and potential investors. CS will attend Angel Networking events and seek out larger scale Venture forums and conferences. We will

communicate our desire to leverage investors' competencies to guide and develop new businesses. Investors will come to associate our business plans and technologies with lowered incurred risks through substantial due diligence and a higher return potential from guided implementation.

We will market to technology inventors by advertising in scientific journals and attending conferences. Additionally, inventors will be compensated for their time and counsel during the evaluation of technologies. We are mindful of faculties' interests, and we value their expertise. We also want to avoid potential conflicts of interest as much as possible by clearly understanding their institution's policy on consulting and commercial interaction.

Sales Plan

CS's sales plan is multifaceted. We will approach sales from both the traditional sales representative standpoint, as well as the more contemporary relationship based sales. For the first quarter of operations, the entire staff will focus on networking into Angel groups, Venture Capital firms, and university technology licensing offices. This approach will yield a large number of potential technologies for evaluation and establish CS as a player in the technology transfer, business development, and investment markets. The initial strategy of all employees interacting with potential customers will continue through the life and culture of CS.

Maintaining and building relationships with customers will be given highest priority for all employees and will allow our services to be continuously reviewed and refined. During years two and three, formal sales personnel will be added to the support team as outlined in the *Prospective Employee* section of this plan. These personnel will concentrate on continuing our relationships with clients and identifying potential investors for technologies in the evaluation and business development processes.

Operations

After performing an initial review of technologies and their potential value to the market, patent attorneys will be retained to perform a review of any patents issued or applications in process by our institutional clients to determine their strengths and weaknesses. We will also assess a technology's potential and the validity of its claimed usefulness through the use of external consultants and industry experts. If we believe the technology is not a one-product company and has strong profit potential, we will approach Angel Investors for funding.

After obtaining Angel investment, we will hire an executive recruitment firm to obtain a CEO, retain a Market Research firm, and other needed consultants to meet the specific needs of the plan. After a complete business plan has been written, it will be presented to Venture Capitalists for first round funding.

Office Considerations

CS will initially operate from leased office space in an academic research hub. The large number of respected, technology-developing universities, as well as venture capital and investment organizations will provide us with a close proximity to our customers. Eventually, we will expand our operations into other technology hotbed markets, such as Boston, Philadelphia, southern California and the Baltimore-Washington D.C. area. After receiving funding, we will investigate incubator opportunities to provide our first office space and support services until our business model is validated.

Supply Considerations

AUTM's membership list is the majority of our proposed institutional customers and technology suppliers. AUTM's web site (<http://www.autm.net>) provides links to many technology licensing offices' homepages, and often, an on-line listing of technologies they have available for licensing. UVentures, Inc. has also expressed a preliminary interest in our services to provide business plans for their clients, comprised primarily of research institutions. After

building strong bonds, we will also look to investors for ideas that they believe are strong but whose current business plans need reworking and further development. After revision, we will resubmit the plan to the referring investor for evaluation.

Process Considerations

Our marketable products are both the business plans we create and the technologies of the companies. The production cycle begins with an initial in-house evaluation of prospective, institutionally developed technologies. The number of technologies evaluated grows proportional to the number of business plan teams. Initial evaluation continues until an adequate number of technologies have been identified for further investigation. Engineers familiar with the technologies' sectors conduct this initial screening. Technologies must satisfy the screening requirements to continue on the commercialization path. These requirements are our initial due diligence and include:

- Product development timeframe.
- Production feasibility.
- Marketability and market size.
- Necessary investment.
- Engineering intuition.
- Substantial return on investment and sustainable profitability.
- Technology can easily be adapted into other products and/or applied to other markets.

The second step of the cycle is a more rigorous in-house evaluation followed by an external expert assessment by leaders in a given technology's field. At this point, technologies are evaluated as either being feasible or infeasible based on more stringent requirements and investor expectations. Funds have been allocated for consulting with the technologies' inventors

as necessary. Technologies that make it beyond the external evaluation are then ranked against one another using the above criteria. The result is a listing of feasible technologies with defined markets and applicable uses.

Immediately following the second step, we approach angel investors with these verified technologies. Having already done much of the due diligence, we will ask angel investors to provide the necessary resources to initiate business development, recruit experienced management, develop a marketing plan, and perform other necessary services. The recruited management CEO becomes a new member of our team and is compensated for four months of work. We provide business development services in-house and everything else necessary to join the technology with venture capital to form a business. If Venture Capital cannot be obtained, we will provide Angels the opportunity to recover some of their investment through licensing fees and/or royalties.

It is important to note that different technologies can be anywhere along the cycle depending on their individual requirements, and this is by no means an annual cycle. CS will constantly watch what each research university is developing and provide appropriate feedback for implementation of the technology in a startup venture.

More importantly, our business concept allows for a business building strategy never before possible. Currently, many institutionally developed technologies cannot be commercialized into a viable business individually. However, by synergistically combining several or many technologies, profitable businesses can be formed. Consider an example from The Pennsylvania State University: “three previously distinct pieces of agricultural intellectual property” were combined together to form a new venture. “Licensed separately, none of the

inventions was likely to amount to much. But united, they formed the basis of an entire new company...”⁹

Team Description

Founding Partners

Eric Molitor received a Bachelors of Science in Chemical Engineering from Bucknell University in 2001. Entrepreneurship is not new to Molitor; he started a network security-consulting firm in 2000. This opportunity provided Molitor with insight into starting a new business. Molitor was also an active member of the Bucknell Entrepreneurs club where he was elected treasurer. Molitor’s interests lie in technology transfer, current events, information technology, and business.

Edward Reilly is a senior undergraduate majoring in business administration at Bucknell University. Reilly has participated in the Institute for Leadership in Technology and Management and was a teaching assistant for an introduction to management course where students form small businesses on campus. During his tenure at Bucknell, Reilly founded and presided as President of the Bucknell Entrepreneurs club. CS’ business concept is an outcome of this club. Reilly’s interests lie in financial services, innovation, strategy, and business.

When evaluating potential business ideas for Bucknell Entrepreneurs, we recognized our lack of access to cutting-edge technologies. To fill this need, we turned to university research and found that technology licensing prevailed greatly over business creation. After identifying this new market niche, CS was born.

We are looking for investors who will assist in finding experienced management to implement our ideas. We intend to be part owners of CS, and co-team leaders during the first year of operation. In subsequent years, we will individually lead each new team through its first year. We realize experience is an important commodity in our proposed venture. We also realize

that seasoned executives may not flock to us simply because of our strategy. Thus, plans are already underway to begin validation of our business model by identifying and commercializing, through a startup, a new university technology. If our belief is true, many technologies sit on the shelf having not obtained a license. If this is the case, we may be able to convince a school to allow us to try to commercialize a technology that was only collecting dust because of their limited resources to marketing it. If we can show some success, it may become easier to recruit experienced employees.

Presently, there are a number of values we find important: interdisciplinary perspectives, creative problem solving approaches, a thoughtful use of available resources, and the ability to have fun. We would want these values to continue into whatever management team is put in place. The descriptions in the following section outline the necessary skills (entrepreneurial, engineering, managerial, business development, market research) recruited personnel must possess as a whole.

Prospective Team Members

CS will foster a collaborative team environment. Technology transfer and business development organizations require extremely knowledgeable and capable employees with a broad understanding of many subjects. When considering compensation requirements, we have budgeted appropriately for the best and brightest individuals. Without this type of employee, CS will have trouble gaining respect from universities and investors alike. As an incentive to attract and retain employees, 30% of CS's minor stakes in portfolio companies will be distributed equally between senior staff at CS, CS founders, and the team members who helped commercialize the technology.

CS seeks to build strong bonds within its company. We plan to have regular team and relationship building sessions, whether it is something as simple as breakfast or a company golf

outing. Being that our company is very team dependent, we feel that relationship development and a fun environment is essential. Coupling Services will be built upon a flat system, initially consisting of one CEO/CFO, a Concierge, and the "Team."

The CEO/CFO will essentially be in charge of the company and step in when there are conflicts to be resolved within the Team. Coupling Services CEO/CFO will possess a minimum of a MBA and will have had at least 10 years of accounting and/or CFO experience, in addition to experience in start-up companies.

The concierge position will be filled by a person with a college degree and a minimum work experience of five years in administrative or HRM areas. The concierge must be very organized, personable and outgoing, for they will be responsible for administrative work, as well as organizing company events.

CS will be employing three engineers, a mechanical, an electrical, and a chemical engineer in the first year. Each of these engineers will have earned a Master's degree in their field and have had a minimum of 5 years experience working with new technology developing or evaluating companies. These engineers will be the primary evaluators of the technologies, due to their deep understanding of the fields. After a technology has been selected as having company startup potential, the engineer's whose field it most closely relates to will join the Team to provide them with technical information necessary, while the others will continue researching and evaluating other new technologies.

The "Team" is the backbone of CS. The Team is responsible for all of the researching, marketing, licensing, and writing of the full business plans. Coupling Services will initially consist of one team composed of a team leader, a marketing professional, a technology transfer specialist, and a business plan specialist.

The team leader must be an individual with extensive technology, entrepreneurial experience. He or she must possess strong leadership skills, be very personable, as well as have a broad technology knowledge base for he or she will be leading the team by keeping it on track and organized. The team leader will deal directly with the technology developer in addition to report directly to the CEO/CFO.

The marketing professional will possess a MBA, with a minimum of ten years experience in technology marketing. They will be responsible for researching all necessary marketing information needed to draw up the business plan.

The business plan specialist will be someone with extensive experience in business development and business plan drafting. They will be primarily responsible for writing the business plan and making sure all the necessary information is included.

The fifth team member, the Technology Transfer expert, must have graduated law school with a Jurors Doctorate and specialized in intellectual law for a minimum of six years. They will be responsible for handling all legal licensing and patenting agreements.

Over the course of four years, CS seeks to add yearly two new teams composed of the same positions. In addition to this, as CS grows, the CEO and CFO positions will develop into two separate entities, with the CFO reporting to the CEO.

The executive recruiting will be done with the help of the venture capitalist, while headhunters and angel investors aid in the seeking out of the remaining positions. We have budgeted for our recruitment needs.

Technology/Business Development Team

Position	Year	Key Responsibilities
Engineer	1-5+	Internal Evaluation and Research Provide Technical Expertise
Marketing Specialist	1-5+	Market Analysis of Tech. Market Plan Creation Internal Evaluation
Business Dev. / Plan Writer	1-5+	Develop Business Models Implementation Plans
Financial Specialist	1-5+	Create Pro Forma Financials Cost Analysis/Requirements
Startup CEO	1-5+	Guide Plan Creation from the Beginning Assist in Obtaining Funding

Administrative / Support Team

Position	Year	Key Responsibilities
CEO/CFO	1	Corporate Vision Spokesperson Management
CFO	3	Treasury Budgeting/Resource Allocation Manage Sales and Marketing
COO	4	Manage and Lead Tenured Teams Control daily Operations
Manager/Engineer	2	Manage Support Staff Infrastructure Perform Tech Development Duties
Technology Transfer Specialist	1	Perform Valuation of Patents, Licenses, and Technologies. Act as a liaison between tech licensing offices and CS

Administrative / Support Team - Continued

Position	Year	Key Responsibilities
Information Technology Guru	2	Provide computer and network support Create Electronic Document Control
Sales Representatives	2&3	Develop relationships with Angels, VCs, and universities. Advocate Technologies and Plans
Research Specialist	2	Aid Teams in research Aid in Documentation Control
Marketing Director	3	Coordinate efforts with marketing firms Provide assistance to Teams Act as a relations coordinator
Concierge/ Receptionist	2	Office Management and employee resource
Engineer	2	Evaluate Technologies Support Teams as needed
Research Assistant	3	Support Research Specialist and Teams

- For the first year, CS will have one team. After our business model is validated, two additional teams will be added each subsequent year.
- Molitor and Reilly will co-lead the first team in year one. In subsequent years, they will individually lead each new team through its first year.

Finances

Financial Highlights

CS requires an initial investment of approximately \$200,000 from an angel investor to fund executive recruitment, market research performed by a respected firm, legal fees including possible intellectual property protection, and compensation and expenses while pursuing venture capital. In return for this investment, Angel investors will have an equity stake in CS. In the first round of funding, we will require three million dollars. This covers all expenses until projected profitability. Alternatively, we project that the validity of our plan will be verified by the conclusion of year one. This verification is our first milestone. If we were to receive milestone based funding, an initial 1.6 million dollars would cover our expenses to this milestone. An additional 1.4 million dollars in second round funding would enable continued operations as projected.

With a three million dollar initial investment, our internal rate of return at the end of five years is 46.9%. Our net present value at the end of five years using a discount rate of 7.5% is \$9.4 million. We provide our services at no cost to the universities supplying us with technologies. Instead, investors are charged a fee, \$50,000 for both angels and venture capital firms who use our services. During the second year (Q6), we will generate a positive cash flow. To fuel future growth and diversify our portfolio, CS will add two additional teams per year following the first year. Pro Forma quarterly financial statements for operations through year five are attached to this plan.

CS will implement an exit strategy after five years in operation. Although an IPO is a possibility, we believe that a company like CS is more likely to participate in a private placement including the investors and institutions that benefit most from our services. Regardless, investors will see our solid history of profitability and our diversified portfolio with equity in a large number of companies as a sound investment.

Key assumptions

The largest assumption is that our goal of funded plans is met each year and there is no need to cut back on production. This assumption is reasonable based on current market projections. Initial Angel investment into the plans we produce, including our fee, will average \$238,000. Angels will fund ten plans in year one, with over 80 technologies undergoing both a thorough internal and external review. Many more technologies will be reviewed internally to feed the 80-technology goal. Other key assumptions include:

- Angels will agree to pay us after we have provided our services.
- The financial figures attached do not include the value of our equity interest in companies we help create. This will significantly increase CS' net worth each year relative to our net income, especially for initial years.

- Our projected cash flows provide us with a cushion against unexpected or underestimated expenses.
- While developing our projected financials, we assume that with time, our production capabilities will improve for more experienced team members. For example, the first time a team writes a business plan, each individual will contribute over 600 hours. At the teams' tenure, the same plan will be developed in a fifth of the time.
- One out of ten plans receiving Angel funding will receive venture capital funding.

A more extensive list of specific financial assumptions is attached to this document.

Equity Distribution

One of the main potential risks we foresee is that CS employees may want to leave CS and join the startup companies they have helped create. This would be very understandable. After our extensive internal and external review process, the opportunities these technologies possess would be compelling. To combat this risk, we have developed an equity distribution model to distribute some of the ownership in these new entities to the workers who helped form them. Please see page 34 for a breakdown of our equity distribution. Employees will now have a clear incentive to stay with CS for the long term. The longer they stay the more equity they have in numerous and diverse companies. We have purposely omitted our negotiation targets for the equity stakes of our angel and venture capital in CS.

Plans for Future Growth

As previously mentioned, many similar technologies can be coupled together to form profitably businesses. This coupling can also reduce costs for administrative and support departments within the business. For example, a large corporation specializing in breast cancer, biotechnology therapies could be formed to house several unique, therapeutic methodologies and products. These similar technologies can share resources, bid collectively on services, and

spread administrative costs between themselves. Typically, many biotechnology startups are based on a small group of narrowly targeted discoveries that are developed into at most a handful of therapies. Without FDA approval, these therapies and the companies they support are worthless. The umbrella strategy we would provide allows for less risk to investors and a possible higher return for successful therapies. Each corporate division would eventually be given a tracking stock or another appropriate form of profit sharing to recognize individual performance within the corporation.

As our business and resources grow, our ability to take on earlier, developmental stage technologies will increase accordingly. Eventually, we will provide monies for developing early stage technologies that meet our thorough review requirements. By doing so, these technologies can more closely complement others already included in our portfolio. This type of technology matching “can ‘synergize’” allowing current ventures to grow.¹¹

We have considered alternative revenue streams to aid in future growth, and as contingency plans if our market projections are unrealized. CS has other distinct revenue options:

- Provide consulting services for portfolio companies: Having worked with the company at the embryonic development stages places us in the perfect position to offer consulting services later to the company.
- If the supply of technologies satisfying our review requirements cannot be met with university developed technology, we will seek to expand into technologies developed for the military by defense supported labs, such as, Milcom Technologies has done.¹⁶
- We have projected a large cash balance at the conclusion of the forth year. This cash will allow us to expand regionally and internationally to explore other markets.

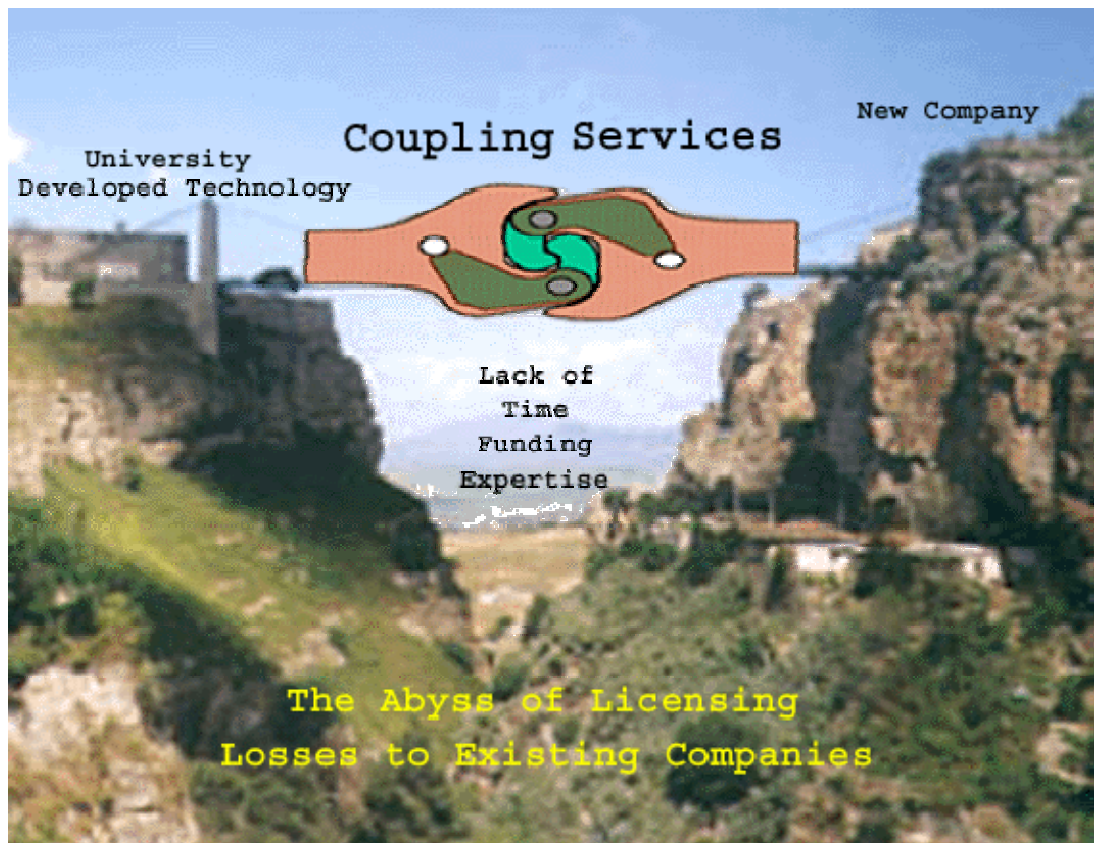
- Take on the role of finding and negotiating licensing agreements for universities with current corporate entities. While reviewing technologies for commercialization, those that do not meet our criteria are rejected back to the university for further licensing considerations. Where applicable, we could aid in the licensing of the technology and recover a larger fee if successful.

Closing Remarks

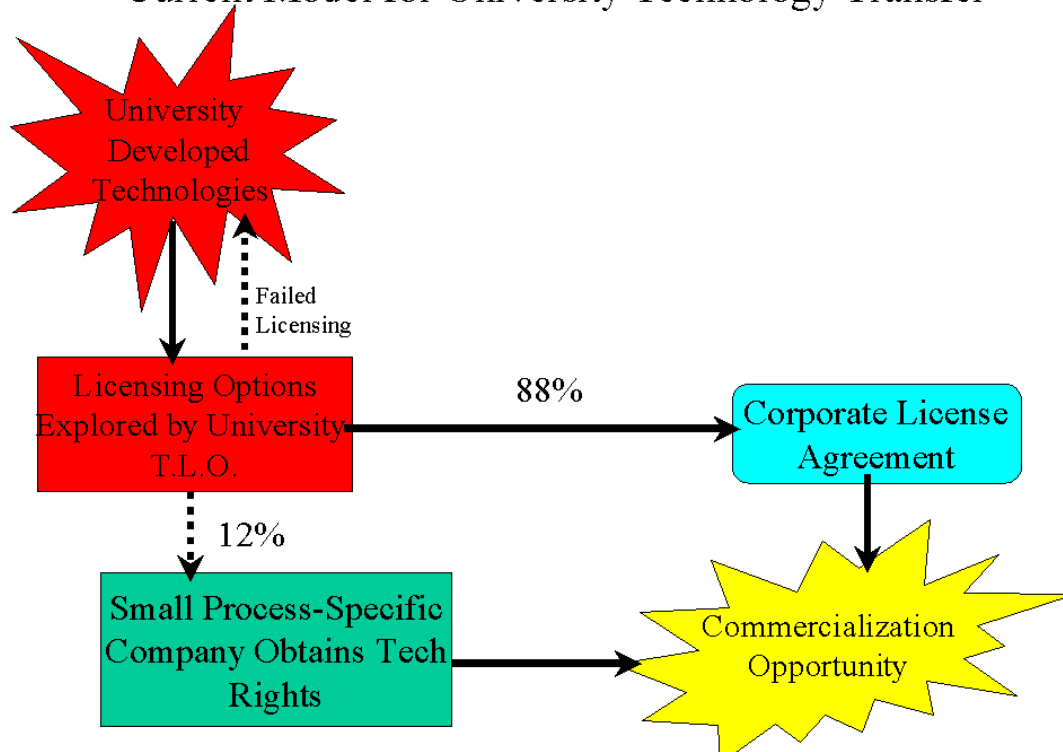
Coupling Services enables technology developing institutions and their researchers to share in the profits of new ventures by taking on the business development role for them leaving them free to do research. We produce business plans with deep industry analysis and critically verified technologies while reducing the risks to investors; we locate start-up capital and bring in experienced management to start and run the new ventures.

If you require additional information or have any comments or suggestions, please visit our website at <http://www.couplingservices.com>, password: bizplaninfo or email: info@couplingservices.com.

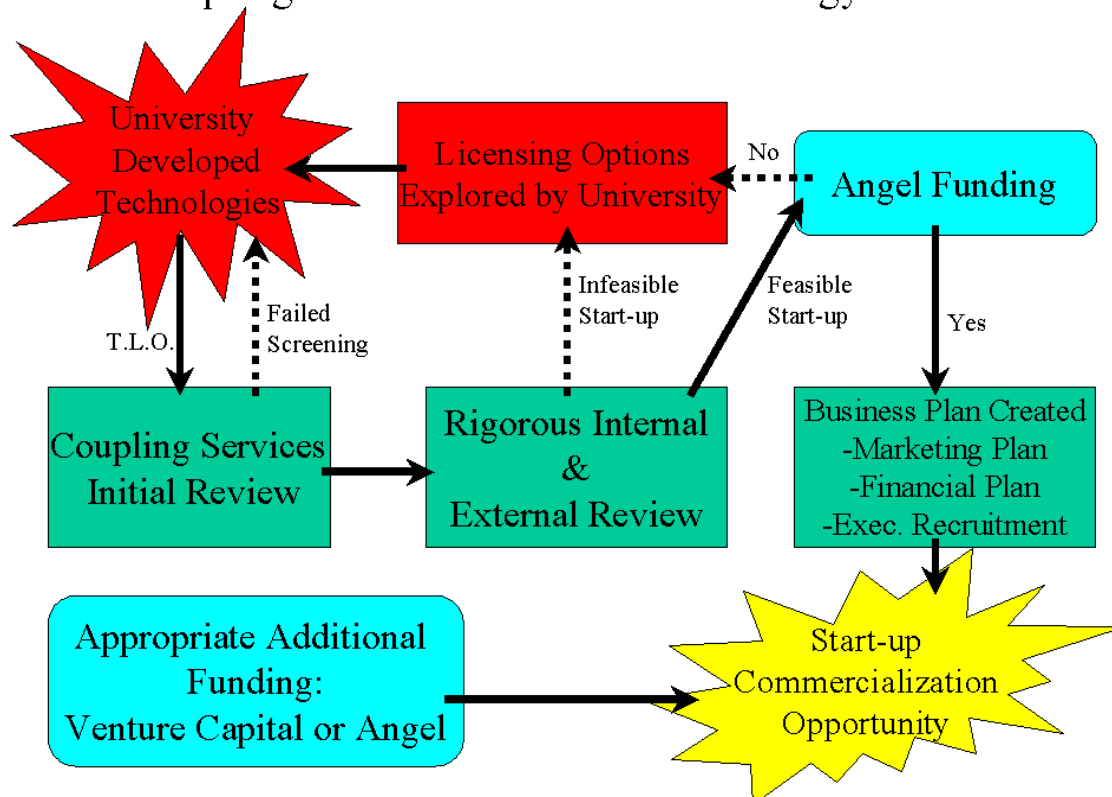
Visual Depiction of Business Concept



Current Model for University Technology Transfer



Coupling Services Model for Technology Transfer



Questionnaire

Dear (name field),

My name is Edward Reilly, and I am a student currently writing a thesis about technology transfer. If you would be so kind as to answer six questions, it would be of great help to me. I would be more than willing to share the results I receive from you and your peers in the industry.

(1) How would you describe the amount of personnel and resources you have to quickly screen and seek licenses for patented technologies? Please place an X by the best descriptive phrase.

- ☐ Severely inadequate amount of personnel and resources
- ☐ Inadequate amount of personnel and resources
- ☐ Adequate amount of personnel and resources
- ☐ More than adequate amount of personnel and resources
- ☐ Abundant supply of personnel and resources

(2) How often do small companies bid against large existing companies for licenses?

- ☐ 0-25% of the time
- ☐ 26-50% of the time
- ☐ 51-75% of the time
- ☐ 76-100% of the time

(3) What percent of what you consider licensable technologies in your portfolio have yet to be licensed to anyone?

- ☐ 0-25% of the portfolio
- ☐ 26-50% of the portfolio
- ☐ 51-75% of the portfolio
- ☐ 76-100% of the portfolio

(4) What percent of the faculty at your institution that have developed a technology could be described as helpful in the licensing process?

- ☐ 0-25% of the faculty
- ☐ 26-50% of the faculty
- ☐ 51-75% of the faculty
- ☐ 76-100% of the faculty

(5) How would you describe the resources your Technology Office has to start new businesses around institutionally developed technology instead of licensing to existing corporation?

- ☐ Severely inadequate amount of resources to start new companies
- ☐ Inadequate amount of resources to start new companies
- ☐ Adequate amount of resources to start new companies
- ☐ More than adequate amount of resources to start new companies
- ☐ Abundant supply of resources to start new companies

(6) Please rank the following factors of choosing a third party service provider for your Technology Office as one being the most important factor and six being the least important factor in choosing a service provider.

- ☐ Location of the service provider
- ☐ The amount of fees the service provider charges
- ☐ The size of the service provider
- ☐ The experience of the service provider
- ☐ Past successes of the service provider
- ☐ Other: _____

If you have any additional comments, feel free to include them in your response. Thank you so much for taking the time to participate in this survey. As soon as the results are tabulated, you will receive a copy.

Milestones

EQUITY DISTRIBUTION MODEL / ORGANIZATION CHART, INITIAL

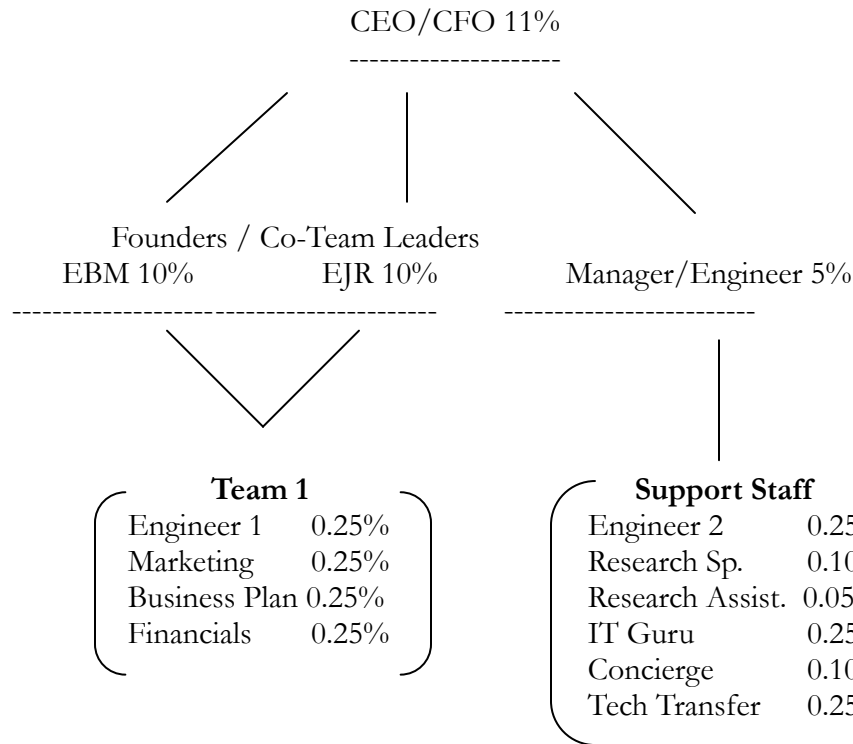
Assumes no turnover

Total Distributed Internally: 38%

Angel Investor(s): ?

Venture Capital Firm(s): ?

Total Available: 62%



EQUITY DISTRIBUTION MODEL / ORGANIZATION CHART, YEAR 5

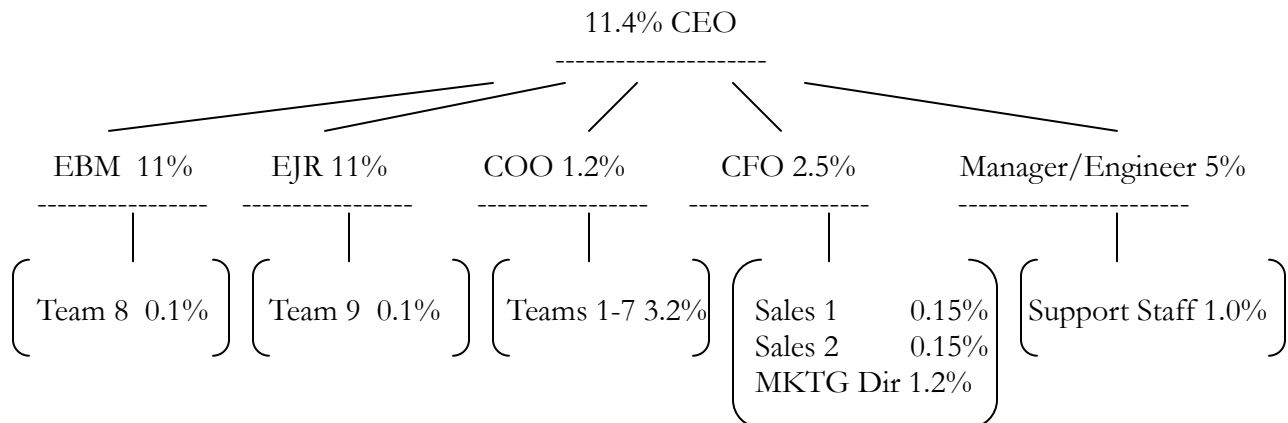
Assumes no turnover

Total Distributed Internally: 48%

Angel Investor(s): ?

Venture Capital Firm(s): ?

Total Available: 52%



Risk Analysis

Potential Risks		Mitigation Plans
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Experienced and successful managers find our business concept and opportunity appealing, but believe that CS' founders' inexperience makes accepting an offer too risky.	By working with our investors to find appropriate management and by providing a highly competitive salary, we will find managers willing to take our vision and ideas and implement them.
By defining a new market niche straddling the technology transfer and business development industries that is not a pure "technology play," may prevent investors who have rigidly defined industry and/or technology parameters from considering CS' plan.	If it becomes obvious through talks with VCs that our plan does not meet their investment criteria because of our defined niche, we will concentrate efforts on obtaining funding from angel groups where industry specific investing is not as common.
Our business model needs a potentially large pool of Angel investors, and it's difficult to predict the likelihood of such a supply and the increasing costs to obtain larger numbers of Angels in the future.	Our sales and marketing strategies with their constant networking approach will by their nature lead us to more potential customers. Also, serial Angels who invest in many opportunities will be valued customers.
Some Angel investors are becoming more like Venture Capitalists and have increasingly stringent "filters" and expectations, including requiring a large portion of experienced management already in place and having current customers before funding.	We will immediately work at establishing customers with universities and investors alike. Each technology will have at least one experienced management person (the future CEO) as part of the plan development team. Many of these filters are overlooked for entrepreneurs without referrals or references, which we will have.
The supply of new technologies appropriate for integration into new businesses that solve problems for customers is not constant or easily predictable.	We will constantly identify high potential technologies and work to develop an available surplus of technologies and universities requiring our services.
Academic institutions choose not to comply with the Bayh-Dole Act and reject our incentives and services. Instead, they choose to license their technologies to large corporations in hopes of a "big winner" and the possible larger and more immediate revenue from royalties of a huge sales volume. ¹⁴	We will offer to sell equity for an initial payment to universities giving them an immediate revenue stream. Also, we will kindly approach compliance issues with universities. Typically, universities will go out of their way to comply with the law.
A significant number of technology developers (professors and researchers) are uncooperative in providing their insights and intimate knowledge of their work, and they discourage or prevent Technology Licensing Offices at their institutions from exclusively licensing technologies to the startups we seek to create.	Most universities have policies in place giving them sole ownership, not the developer, of any technology created there. For most technologies more than one developer is involved. For instance we will heavily recruit graduate students to provide knowledge into technologies they have worked on, offering them roles within the startup if possible.
Currently, there are low barriers to entry for competition, and CS may not be approved for intellectual property protection quickly or at all in the current business process patent backlash.	We hope to be the first to market with this concept giving us the recognized success and credibility. Also, being the first to develop contracts with universities can give us an edge and allow us the first chance to evaluate and commercialize new technologies.
CS' profitability is limited until several teams are added, which defray overhead expenses.	The first year of operations will further validate our concept. We will add 2 teams each subsequent year.
The timing of our cash inflows (i.e. when Angels fund our efforts) is very uncertain while our outflows are fairly constant, and too large of an imbalance could render CS illiquid.	If cash flows become a concern, we will focus our attention on technologies further along in development with a shorter time to market. Also, once established, we will obtain a revolving line of credit.
The decision to not initially focus on the biotech industry may limit our supply of technologies so much that we are unable to reach our projections in the number of companies funded.	Biotech will never be an excluded technology industry. If necessary, or if enough interest in terms of time and capital are available, we will pursue biotechnologies. Most biotech requires much time and capital.
It is difficult to successfully combine the three elements of business/industry expertise, technical competency, and converting a technology into viable, profitable products.	By providing sufficient compensation for the best and brightest employees, we greatly reduce this risk through implementation management.

Five-Year Pro Forma Balance Sheet in Thousands of Dollars

Balance Sheet	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20
Assets																				
Current Assets																				
Cash	2484	2002	1677	1516	759	886	1094	1946	1781	2506	3488	5246	5702	7144	8833	11538	12760	15008	17494	21236
Accounts Receivable		251	487	981	1917	2327	3352	3762	4494	5314	6134	6544	7275	8095	8915	9325	10057	10877	11697	12107
Total Current Assets	<u>2484</u>	<u>2253</u>	<u>2163</u>	<u>2497</u>	<u>2677</u>	<u>3214</u>	<u>4447</u>	<u>5708</u>	<u>6275</u>	<u>7820</u>	<u>9622</u>	<u>11790</u>	<u>12978</u>	<u>15240</u>	<u>17749</u>	<u>20864</u>	<u>22817</u>	<u>25885</u>	<u>29191</u>	<u>33343</u>
Land, Buildings, and Equipment																				
Land, Buildings, and Equipment	71				85				64				80				53			
Less Accumulated Depreciation	4	4	4	4	8	8	8	8	11	11	11	11	15	15	15	15	18	18	18	18
Net Land, Buildings, and Equipment	<u>68</u>	<u>64</u>	<u>61</u>	<u>57</u>	<u>135</u>	<u>127</u>	<u>119</u>	<u>111</u>	<u>164</u>	<u>153</u>	<u>142</u>	<u>131</u>	<u>196</u>	<u>181</u>	<u>166</u>	<u>151</u>	<u>185</u>	<u>168</u>	<u>150</u>	<u>132</u>
Total Assets	<u>2551</u>	<u>2317</u>	<u>2224</u>	<u>2554</u>	<u>2811</u>	<u>3340</u>	<u>4565</u>	<u>5819</u>	<u>6438</u>	<u>7972</u>	<u>9763</u>	<u>11920</u>	<u>13173</u>	<u>15420</u>	<u>17914</u>	<u>21014</u>	<u>23003</u>	<u>26053</u>	<u>29341</u>	<u>33476</u>
Liabilities																				
Current Liabilities																				
Accounts Payable, Plan Related		110	220	440	770	990	1540	1760	1980	2420	2860	3080	3300	3740	4180	4400	4620	5060	5500	5720
Accounts Payable, Other	80	80	80	80	80	80	80	80	55	55	55	55	55	55	55	55	55	55	55	55
Income Tax				39		108	274	458	309	491	645	904	678	870	1023	1366	1097	1297	1451	1878
Total Current Liabilities	<u>80</u>	<u>190</u>	<u>300</u>	<u>559</u>	<u>850</u>	<u>1178</u>	<u>1894</u>	<u>2298</u>	<u>2344</u>	<u>2966</u>	<u>3560</u>	<u>4039</u>	<u>4033</u>	<u>4665</u>	<u>5258</u>	<u>5821</u>	<u>5772</u>	<u>6412</u>	<u>7006</u>	<u>7653</u>
Stockholders' Equity																				
Capital Investment at Year Zero	3000																			
Retained Earnings (Loss)	-529	-344	-204	72	-34	201	509	850	573	912	1197	1678	1259	1615	1900	2537	2038	2409	2695	3488
Total Stockholders' Equity	<u>2471</u>	<u>2127</u>	<u>1924</u>	<u>1996</u>	<u>1961</u>	<u>2162</u>	<u>2671</u>	<u>3521</u>	<u>4095</u>	<u>5006</u>	<u>6204</u>	<u>7882</u>	<u>9141</u>	<u>10755</u>	<u>12656</u>	<u>15193</u>	<u>17231</u>	<u>19640</u>	<u>22335</u>	<u>25823</u>
Total Liabilities and Equity	<u>2551</u>	<u>2317</u>	<u>2224</u>	<u>2554</u>	<u>2811</u>	<u>3340</u>	<u>4565</u>	<u>5819</u>	<u>6438</u>	<u>7972</u>	<u>9763</u>	<u>11920</u>	<u>13173</u>	<u>15420</u>	<u>17914</u>	<u>21014</u>	<u>23003</u>	<u>26053</u>	<u>29341</u>	<u>33476</u>

Five-Year Pro Forma Income Statement and Statement of Cash Flows in Thousands of Dollars

Income Statement	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20	
Revenue		251	512	1066	2017	2502	3577	4342	4894	5764	6684	7664	7975	8845	9765	10985	11057	11927	12847	14307	
Operating Expenses		525	591	712	952	2044	2185	2786	3027	4001	4350	4831	5071	6024	6346	6827	7067	7905	8203	8683	8924
Depreciation		4	4	4	4	8	8	8	8	11	11	11	11	15	15	15	15	18	18	18	18
Operating Income		-529	-344	-204	110	-34	309	783	1308	882	1403	1842	2582	1936	2485	2924	3903	3135	3707	4146	5366
Income Tax Payable					39		108	274	458	309	491	645	904	678	870	1023	1366	1097	1297	1451	1878
Net Income		-529	-344	-204	72	-34	201	509	850	573	912	1197	1678	1259	1615	1900	2537	2038	2409	2695	3488

Cash Flow Statement	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20
Cash Flow from Operations																				
Net Income	-529	-344	-204	72	-34	201	509	850	573	912	1197	1678	1259	1615	1900	2537	2038	2409	2695	3488
Accounts Receivable		-251	-236	-494	-937	-410	-1025	-410	-732	-820	-820	-410	-732	-820	-820	-410	-732	-820	-820	-410
Depreciation Expense	4	4	4	4	8	8	8	8	11	11	11	11	15	15	15	15	18	18	18	18
Income Tax Expense				39	-39	108	166	184	-149	182	154	259	-226	192	154	343	-269	200	154	427
Accounts Payable, Other	80								-25											
Accounts Payable, Plan Related		110	110	220	330	220	550	220	220	440	440	220	220	440	440	220	220	440	440	220
Total From Operations	<u>-445</u>	<u>-481</u>	<u>-326</u>	<u>-160</u>	<u>-672</u>	<u>127</u>	<u>208</u>	<u>852</u>	<u>-102</u>	<u>725</u>	<u>982</u>	<u>1758</u>	<u>536</u>	<u>1442</u>	<u>1689</u>	<u>2705</u>	<u>1275</u>	<u>2247</u>	<u>2486</u>	<u>3742</u>
Cash Flow From Investing																				
Land, Buildings, and Equipment	-71				-85				-64				-80				-53			
Total From Investing	<u>-71</u>				<u>-85</u>				<u>-64</u>				<u>-80</u>				<u>-53</u>			
Cash Flow From Financing																				
Capital Investment at Year Zero	3000																			
Total From Financing	<u>3000</u>																			
Net Change In Cash	-516	-481	-326	-160	-757	127	208	852	-165	725	982	1758	456	1442	1689	2705	1222	2247	2486	3742

Profitability Measures:

IRR (5 years)	46.89%
NPV (5 years) Using Prime Interest Rate of 7.5%	\$9,409,085

Select Financial Assumptions: Complete assumptions available upon request.

Tax Rate	0.35	Telephone	
Hourly Cost For Work by Team Member	75.00	Base Telephone / Employee	91.88
Hours Worked Per Week / Employee	50	Time spent on phone (%)	0.05
Weeks Worked Per Quarter	12.25	Cost per minute	0.05
Average Travel Expenses / Tech	10000	Base Telephone / Tech	4.29
CS Travel Expenses		Time spent on phone (%)	5.00
Base Travel Expenses / Quarter	20000	Cost per minute	0.05
Travel Expense / Employee / Quarter	7000	Overall Base Telephone	2000.00
Average Cost of External Tech Assess	15000	Internet	
Average Cost of Exec Recruit /Tech	20000	Broadband to office / Quarter	800.00
Initial CEO Salary /Tech	50000	Individual Access / Quarter	480.00
Inventor Consulting Cost /Tech	3000	Misc Overhead / Tech	10000
Average Legal Fee / Tech	12000	Marketing Plan Cost (External) / Tech	10000
CS Fee to Angel / Tech	50000	Initial Review Cost / Tech	25000
CS Fee to VC-Startup / Tech	50000	Lease in Year One	72000
Plans VC Funded / Plans Angel Funded	0.10	Human Resources yearly cost	50000
CS Fee if Licensed	25000	Marketing Firm (CS) quarterly cost	37500
UMR Assumed rate function(lease payment)	0.35	Year 1 -2	150000
Postage / Quarter		Year 3+	50000
Base Postage / Employee	100.00	Insurance yearly cost	50000
Base Postage / Tech	200.00	Accounting Firm yearly cost	70000
Overall Base Postage	500.00		

Hiring/Salary Assumptions

Sample team shown. Two additional teams added per year.

Production/Market Assumptions

	Year	1	2	3	4	5
Total Plans Produced / Year	7	46	94	142	190	
Market Demand Projections	273	383	393	403	413	
Estimated Market Share (%)	2	12	24	35	46	

Salaries per Year	Year 1	Year 2	Year 3	Year 4	Year 5
Average Raise Per Year	4.00%				
Average Benefit Package % of Salary	40.00%				
CEO/CFO	220000	228800	237952	247470	257369
EBM	50000	52000	54080	56243	58493
EJR	50000	52000	54080	56243	58493
CFO			180000	187200	194688
COO				180000	187200
Sales Rep 1		100000	104000	108160	112486
Sales Rep 2			100000	104000	108160
Information/Research Specialist		30000	31200	32448	33746
Marketing Director			180000	187200	194688
Concierge/Receptionist		30000	31200	32448	33746
Research Assistant		25000	26000	27040	28122
Manager / Engineer 2		90000	93600	97344	101238
Engineer 3		90000	93600	97344	101238
Technology Transfer Specialist	45000	46800	48672	50619	52644
Information Technology Guru		60000	62400	64896	67492
Sample Team Members					
Engineer 1	90000	93600	97344	101238	105287
Marketing Specialist	100000	104000	108160	112486	116986
Business Dev/Plan Writing Specialist	130000	135200	140608	146232	152082
Financial Specialist	100000	104000	108160	112486	116986

Budget Analysis

Overview

- Similar to most service based organization, the focus of our first year budget is on recruiting and paying our employees. The budget allocates \$1.5 million for these activities. This focus on our employees will allow us to find the best and most experienced in our niche industry.
- Outside of salary expenses, the budget focuses on our need to quickly obtain clients, build relationships, and validate technologies. Almost \$500 thousand have been allocated for first year travel expenses. This will allow us to travel to the birthplace of emerging technologies and more easily evaluate their commercialization potential.
- An additional \$350 thousand is allocated for use in obtaining experienced management for the technologies found to meet our stringent requirements.
- The balance of our budget is for operational expenses. The large expenditures include marketing, executive recruitment, and technology verification.
- All years after the first year are scaled appropriately for the number of employees and anticipated additional expenses related to continuing the business.

Summary Statement of Transaction, Budget Year One, In Thousands of Dollars				
	Year One	Year Two	Year Three	Year Four
Revenues	1,881	12,784	25,711	38,637
Program Spending	3,547	10,093	18,306	26,319
Investment	3,000	--	--	--
Budgetary Balance	1,334	2,691	7,405	12,318

Our budget is a moderately cautious document that is designed to maximize the return to our investors while providing amply resources to every required item.

References Cited

1. Pressman, Lori. AUTM licensing survey summary: fiscal year 1999. Northbrook, IL: AUTM (Survey, Statistics, and Metrics Committee), 2000.
2. Himmelstein, Linda. "Crunch Time for VCs," Business Week, February 12, 2001: 23-32.
3. Lytton, Michael. "Advice of Counsel, Q&A On Tech Transfer" at Palmer & Dodge LLP, <http://www.palmerdodge.com/about/publicationoutput.cfm?fileID=9>
4. "University licensing of technology to industry has grown markedly over the past decade," Chemical and Engineering News, August 24, 1992: v70 n34 p17.
5. Webster, Andrew. "University-corporate ties and the construction of research agendas," Sociology, February 1994: v28 n1 p123(20).
6. Byrne, Noel. Licensing technology: drafting and negotiating agreements. New York: Macmillan Press, 1994.
7. Nelsen, Lita. "Increase of Intellectual Property Licensing at Universities Stems from Changes in Funding and Legislation." MIT Tech Talk, August 26, 1998. <http://web.mit.edu/newsoffice/tt/1998/aug26/nelsen.html>
8. "Angle To Set Up New U.K. Companies," Genetic Engineering News, April 15, 1999: v19 n8 p56.
9. Zacks, Rebecca. "The University Research Scorecard," Technology Review, July/August 2000: v103 n4 p88-90.
10. "University-affiliated Venture Capital Funds," Health Affairs, Summer 1994: v13 n3 p159.
11. Tornatzky, Louis G, et al. Benchmarking Best Practices for University-Industry Technology Transfer: Working with Start-Up Companies, October 20, 1995: p10.
12. Governmental Relations, The Council on. "University Technology Transfer, Questions and Answers," November 30, 1993.
13. Thayer, Ann. "Closing the Venture Gap," Chemical & Engineering News, August 14, 2000: v78 n33 p 17-20.
14. Tornatzky, Louis G, et al. Benchmarking University-Industry Technology Transfer in the South: 1995-1996 Data, July 1997 p11.
15. Siegel, Bari Faye. "Transfer Technology to Boost Your Business," <http://startup.wsj.com/runbusiness/survival/200012111008-siegel.html>
16. Squeo, Anne Marie. "Secret weapons: High-Tech Prospector Digs for Gold in Troves of the Defense Giants—Mike Buffa Finds Promise In Notions Big Firms Had Little Idea How to Market." Wall Street Journal. April, 25, 2001: p. A1.
17. "About Science and Technology Ventures," <http://www.columbia.edu/cu/cie/about.html>
18. "A Review of University Industry Research Relationships," http://www.upenn.edu/ctt/about_tech/u-i_review.shtml

Where Products Sit on Shelves

An Initial Identification of Target Markets

A Coupling Services' Report
Prepared by Eric Charles
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Department of Psychology
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Summary

In the months of January and February 2002, Coupling Services sent a six question survey to licensing officers of every college and university based technology licensing center within the United States. The questions were designed to determine both the adequacy of current staff levels, the perceived demand for a business such as Coupling Services, and the opportunity to create demand. The results show that there is an identifiable niche for Coupling Services, they suggest an identifiable initial target market, and suggest the most successful avenues by which to approach that market. Specifically, the initial customer base should be small to mid-sized colleges and universities, which tend to think they are inadequately staffed, have an overabundance of licensable technologies, and rarely if ever have small companies competing to gain licenses. The primary quality of Coupling Services, which should be emphasized when approaching these companies, is that there is no direct cost to the school, and evidence suggests that the company will be unhindered by either its small size or distant location.

Please note: The complete data set as well as further elaboration upon the statistical methodologies that lead to any of the conclusions stated herein is available upon request.

Methodology

A list of the 116 technology licensing offices based in colleges and universities in the United States was compiled with an office member selected to receive the survey (Appendix 1). As the surveys will be used in part to complete Edward Reilly's thesis on technology transfer, they were advertised as such to encourage a stronger return rate. Out of those solicited, ten were no longer at the office leaving a potential response rate of 106. Thirty-one responses have been received at this point yielding a sample size that is almost 30% of the population to which the results will be generalized. The results of the survey were combined with already gathered data as to both the size of the College/University and the size of its technology office. The analysis consisted of identifying the descriptive parameters of each variable, and determining correlations between variables. Results that reported as significant will have passed at least a .05 alpha level and any other interesting results will be clearly labeled as "trends".

Results

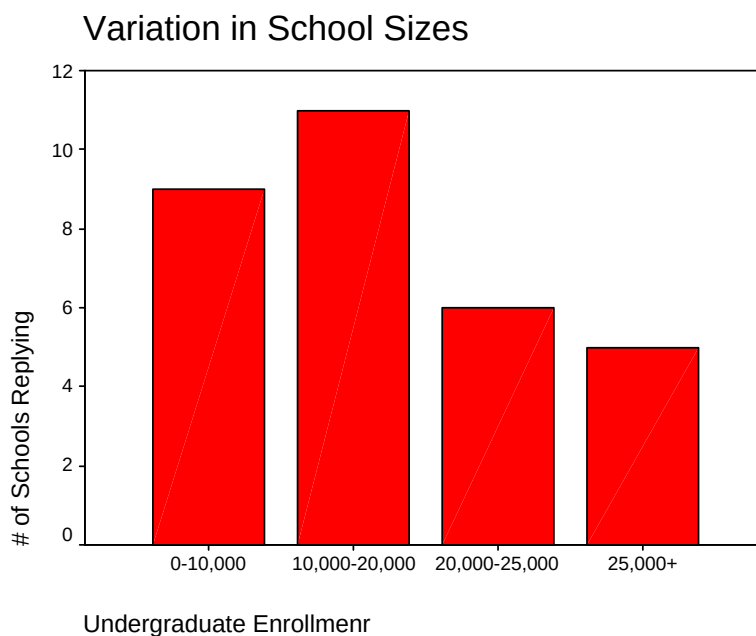
Descriptive Statistics

Public vs. Private

Of the schools which replied, 23 were public, 8 private.

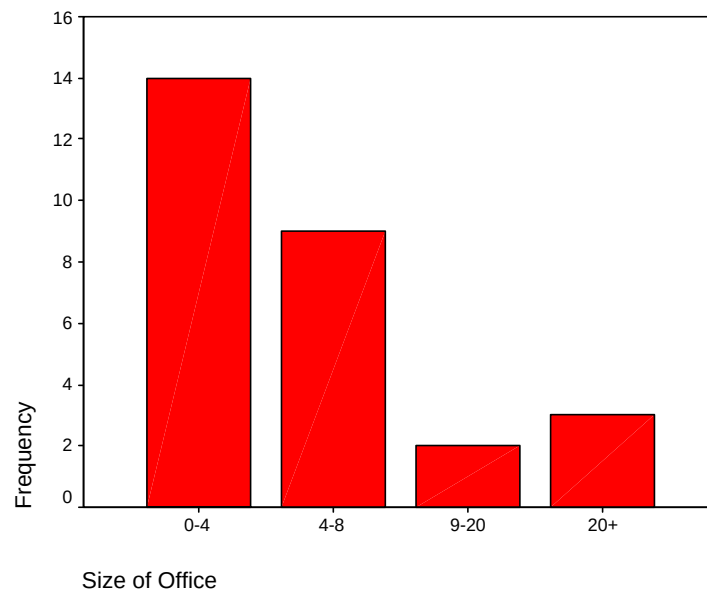
Size of School

The sizes of replying schools (based on undergraduate enrolment) ranged from 0 at The Rockefeller University, which is a graduate student only medical research school to 34,406 at Penn State. Due to the unusual circumstances surrounding The Rockefeller University, which made it an obvious outlier in most analyses, it was excluded from correlational analysis. For these analyses, Rice University, with 2,658 undergraduates enrolled, was the smallest university.



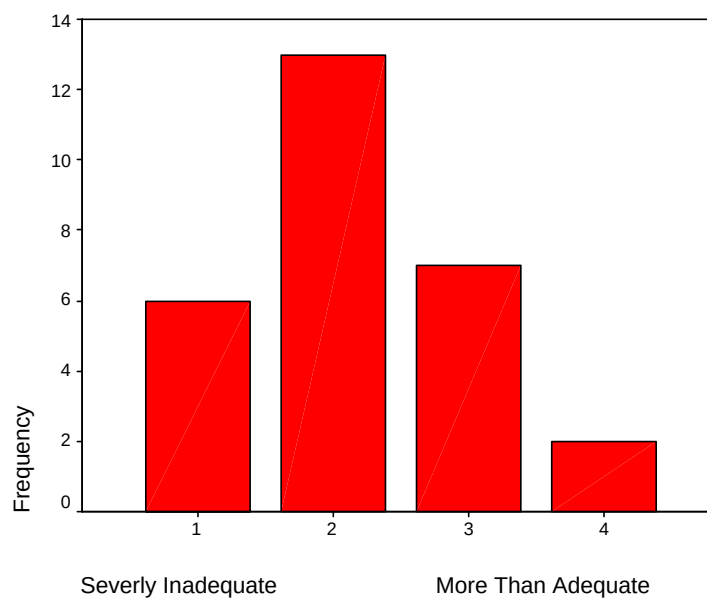
Size of Licensing Office

The average size of a technology licensing office was 7.5 people. This was inflated by a few outliers, specifically the University of Wisconsin and the University of Michigan which both had more than 30 employees. Half of those who responded had less than 4 fellow employees.



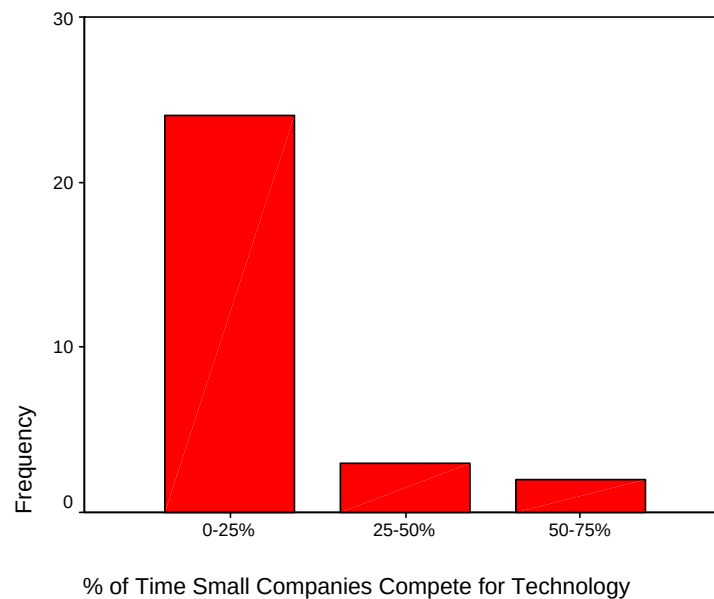
Question 1: How would you describe the amount of personnel and resources you have to quickly screen and seek licenses for patented technologies?

The majority of companies (68%), which responded to this question, reported that their resources were either inadequate, or severely inadequate. Only 7% claimed to have more than adequate resources.



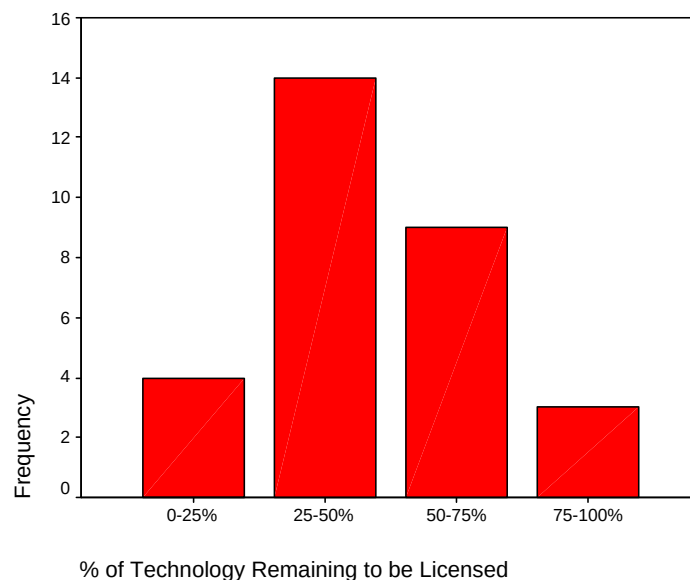
Question 2: How often do small companies bid against large existing companies for licenses?

Most Schools reported little if any competition by small companies to license their resources. Only 7% of schools reported that competition happened the majority of the time.



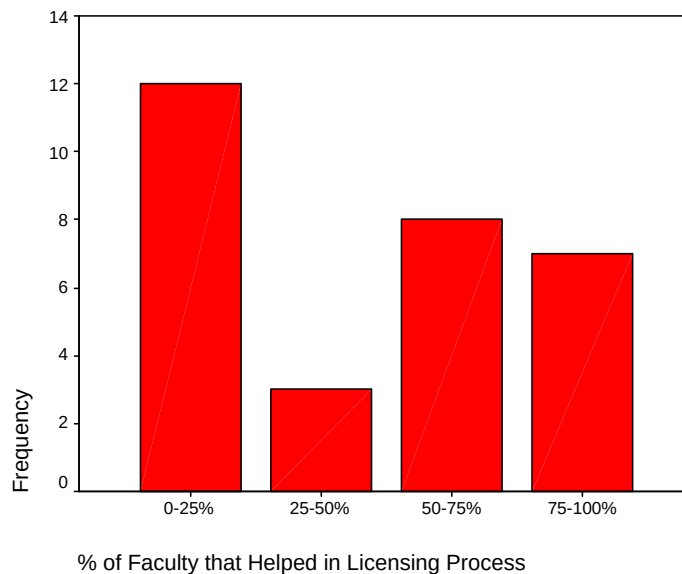
Question 3: What percent of what you consider licensable technologies in your portfolio have yet to be licensed to anyone?

Forty percent of those who replied reported that more than half of their licensable technologies are not being taken advantage of. More than 75% of the remaining schools reported that between a quarter and half of their technologies remain only on the books.



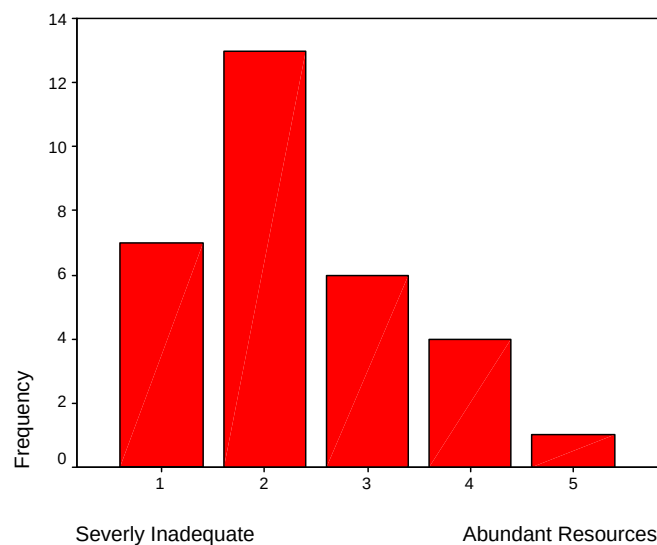
Question 4: What percent of the faculty at your institution that have developed a technology could be described as helpful in the licensing process?

While the most common answer indicated that 40% of licensing offices get very little help from faculty whose technologies they are trying to license, many schools are getting at least minimal help from quite a large percentage of faculty members.

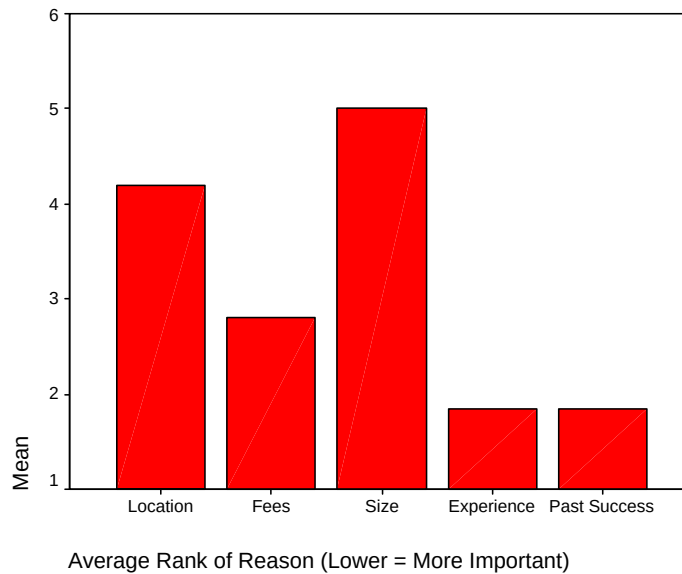


Question 5: How would you describe the resources your Technology Office has to start new businesses around institutionally developed technology instead of licensing to existing corporation?

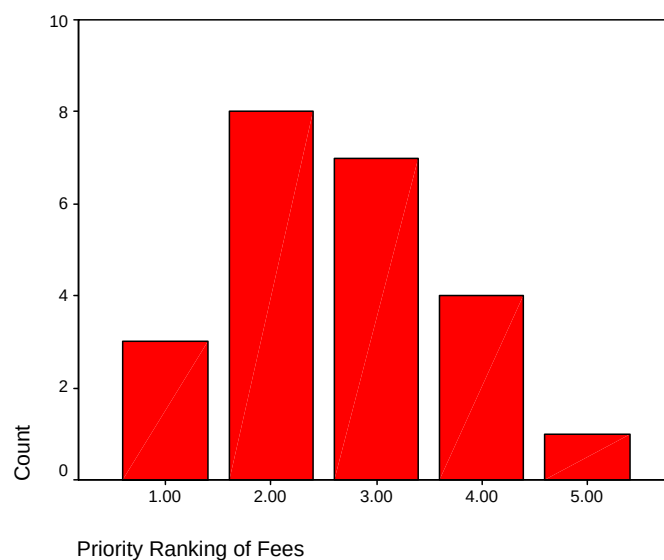
Two-thirds of respondents reported their resources for starting a new company to be inadequate, or worse. Twenty percent described their resources as more than adequate.



Question 6: Please rank the following factors of choosing a third party service provider for your Technology Office as one being the most important factor and six being the least important factor in choosing a service provider.

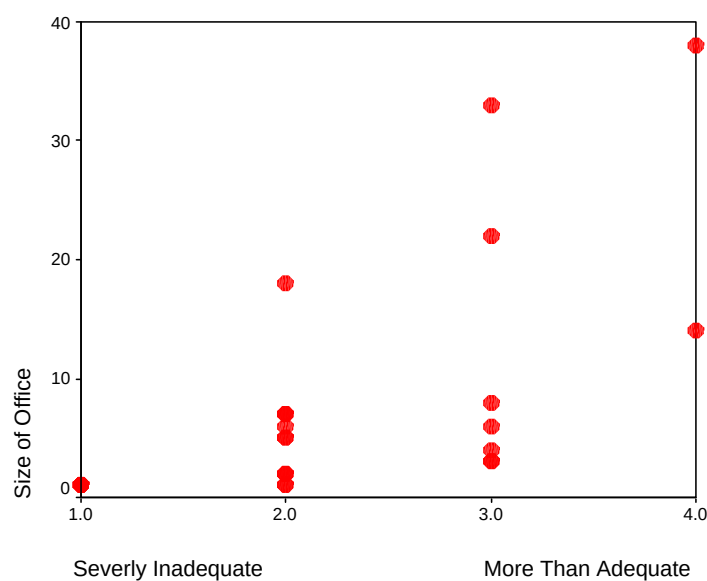


The priority Ranking of Fees will be highlighted separately, as one advantage to Coupling Services' business design is that there are no fees charged to the college or university. Almost half of respondents (47%) ranked the amount of fees as either their number one or number two priority.



Correlations

The most significant correlation was between office size and office adequacy ($r = .615$, $N = 24$, $p = .001$). While this result may be driven primarily by the larger offices (those with over 15 employees), 81% of offices to which inquiries were sent had less than 15 employees. This means that approximately 40% of all technology licensing departments can be assumed to feel inadequately staffed if we have a representative sample.



Other significant relationships included that those offices which considered themselves to be poorly staffed also reported inadequate resources to start a new business based on their technologies ($r = .471$, $N = 27$, $p = .013$). Those schools, which considered themselves less prepared to start a new business, reported a higher percentage of potential technologies that were not being taken advantage of ($r = .382$, $N = 29$, $p = .041$). Larger schools in the sample tended to be public schools ($r = .368$, $N = 30$, $p = .046$) and larger schools tended to have larger technology licensing offices ($r = .422$, $N = 27$, $p = .028$). These schools, which already had increased resources, also reported that a higher percentage of faculty members helped during the licensing process ($r = .385$, $N = 29$, $p = .039$).

Conclusions

The typical respondent was from a public college with approximately 17,000 students. They felt their office, with 8 staff members, had an inadequate amount of personnel, that a significant number of their licensable technologies had not been licensed, and that they had inadequate resources to attempt to develop a new business around the technologies they had. They rarely, if ever, had small companies competing with large corporations for their products and received help from less than half of the professors whose products they were trying to license. While they tended to prefer service providers that had past experience and proven successes, they put a strong priority towards the fees charged by the provider, while caring little about the size of the company or where it was located.

Correlations in the data indicate that very large public schools tend to highly staffed offices, receive a large amount of faculty support in the licensing process, and feel they have adequate resources to start a new business around their technologies. In contrast, small to mid-sized universities tend to think they are deficient in at least one of the areas surveyed.